

Tide retrospective: the second-order J_n expansion (gamma-rung programme, stage 2b)

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Stage 2a built the cubic-order remainder layer; this tide completes stage 2 with the exact algebra and the headline. Together they upgrade the seabed's rescaled integral asymptotics to second order, which is the analytic heart of the gamma-rung programme: the $A^2/(2t)$ coefficient made visible here is what survives into the fourth cumulant and carries α^2/λ^5 against γ/λ^4 .

2 The thing formalised

Three declarations in `Laplace/OneD/JnSecondOrder.lean`, sorry-free with zero warnings, plus a one-word de-privatisation of the seabed's J_n -integrand integrability. The exact decomposition¹: for $t > 0$, with A, B the cubic and quartic scales and $m_k = \int u^k e^{-u^2/2}$,

$$\int u^n e^{-u^2/2} \left(1 - s_t + \frac{s_t^2}{2}\right) = m_n - \frac{A}{\sqrt{t}} m_{n+3} - \frac{B}{t} m_{n+4} + \frac{A^2}{2t} m_{n+6} + \frac{AB}{t\sqrt{t}} m_{n+7} + \frac{B^2}{2t^2} m_{n+8},$$

and the stage headline²:

$$J_n(t) = m_n - \frac{A}{\sqrt{t}} m_{n+3} - \frac{B}{t} m_{n+4} + \frac{A^2}{2t} m_{n+6} + O\left(\frac{1}{t\sqrt{t}}\right) \quad (t \geq 1),$$

with the AB and B^2 cross-terms absorbed into the explicit error constant. The numerical check (executed, archived) confirms the error scale with room at $n = 2$.

3 Proof strategy

The pointwise six-term identity is where the new difficulty lived: the square s_t^2 mixes $1/\sqrt{t}$ and $1/t$ denominators, and `ring` does not know $(\sqrt{t})^2 = t$. The resolution is the *$\sqrt{t}$ -atom substitution*: after `set st := Real.sqrt t` folds every $\sqrt{t}$ in the goal into the opaque local `st`, the remaining bare t 's are rewritten by $t = st \cdot st$ — safe precisely because the substitution target no longer contains $\sqrt{t}$ syntactically — and the identity becomes rational in `st`, closed by `field_simp` and `ring`. The integral split is a six-fold nested `integral.add` with type-ascribed cumulative witnesses per the repo's `Pi.add` discipline; the headline then splits J_n into the quadratised part plus the stage-2a remainder and routes the two cross-terms into the constant via $t^2 \geq t\sqrt{t}$.

¹`quadratised_integral_decomposition.`

²`Jn_asymptotic_order2.`

4 Roadblocks and resolutions

Four build iterations. The one instructive failure: an `Integrable.congr` whose pointwise goal contained an unapplied `Pi.sub` — the fourth member of the repo’s `Pi`-form family (`add`, `div`, derivative `sub/sum`, and now integrand `sub`), with the same one-line repair. Two further iterations removed tactics that `gcongr` had already made redundant.

5 What was Mathlib, what was new

Mathlib: integral linearity, `gcongr`, the triangle inequality. Seabed: the linearised decomposition supplied the assembly pattern; the Gaussian-moment integrability family and stage 2a supplied every analytic input. New: the $\sqrt{t}$ -atom substitution, promoted to the repo `CLAUDE.md` as the standard way to prove rescaled-time polynomial identities.

6 Lessons

When an algebraic identity mixes t and $\sqrt{t}$, fold the radical into an atom *first* and rewrite t afterwards; the substitution is only unsafe while the radical is still spelled with t inside. First-attempt success on both the identity and the six-fold split shows the catalogue disciplines have compounding value: this tide’s genuinely new content compiled clean, and every failure was a known class.

7 Follow-ups

- Stage 3: second-order moment-ratio lemmas (the Gibbs moments $\langle x^r \rangle = (\lambda t)^{-r/2} J_r / J_0$ expanded through order $1/t$ relative to leading).
- Stages 4–6 per the programme plan (κ_4 assembly, the analytic limit, gamma recovery).